Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Physics

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Work, Energy and Power

DM

Unit 1: Work, Energy and Power

Learning Objectives

• Students will define work, energy and power and state their SI units.

• Students will calculate work done by a force and distinguish kinetic energy from potential energy.

• Students will apply the idea of energy conservation to simple everyday examples.

Key Concepts

• Work is done when a force moves an object in the direction of the force; work = force x distance, measured in joules (J).

• No work is done if there is no movement, or if the force acts at right angles to the movement, such as carrying a bag while walking on level ground.

• Energy is the capacity to do work; its SI unit is also the joule.

• Kinetic energy is the energy of a moving object and depends on its mass and speed: KE = 1/2 mv^2.

• Potential energy is stored energy due to position, such as the gravitational potential energy of a raised object: PE = mgh.

• Power is the rate of doing work, measured in watts (W); one watt equals one joule per second.

Teaching Methodology (Step by Step)

1. The teacher pushes a desk across the room and asks when work is done and when it is not, introducing the definition of work.

2. The teacher works out the work done in lifting a 5 kg bag through 2 m and shows the unit joule.

3. The teacher rolls a ball and a heavier ball down a ramp to show that faster and heavier objects carry more kinetic energy.

4. The teacher holds a stone above the floor and releases it, explaining how stored potential energy changes into kinetic energy.

5. The teacher derives the power formula with the example of two students climbing stairs at different speeds.

6. The teacher asks students to identify energy changes in a swinging pendulum and a falling coconut.

Classroom Activities

• Work hunt: students list five daily actions and decide in each case whether work is done in the physics sense.

• Energy race: in pairs, students calculate the kinetic energy of a 2 kg ball moving at 3 m/s and of a 4 kg ball moving at 2 m/s and compare them.

• Power check: students estimate the power of a 60 W bulb in one hour in joules and discuss what the number means.

Assessment Questions

1. A force of 20 N pushes a box 5 m. How much work is done?

2. A 10 kg stone is held 3 m above the ground. Find its gravitational potential energy (g = 10 m/s^2).

3. Explain the energy changes when a ball is thrown up and then falls back down.

Homework

• Calculate the work done in carrying a 15 kg school bag up 4 m of stairs and the power if it takes 20 seconds.

• Write three examples each of kinetic energy and potential energy from your home or school.

Time Allocation

40 minutes

DM